DIYA SHETTY

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EDUCATION

North Carolina State University – Raleigh, NC *May 2026*
Master of Computer Science (MCS) | GPA: 4.0/4.0
Coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Foundations of Data Science, Computer Networks

BMS College of Engineering – Bengaluru, India *May 2024*
Bachelor of Engineering (B.E.) – Information Science and Engineering

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, SQL, C++, HTML, CSS, R
- **AI/ML Engineering:** LangChain, RAG, Agentic AI, LLM orchestration, vector search (ChromaDB), prompt engineering, TensorFlow
- **Frameworks & Libraries:** Spring Boot, Spring Security, FastAPI, Next.js, React.js, Node.js, Pandas, NumPy
- **Cloud & Databases:** AWS (Glue, S3), Azure, PostgreSQL, MySQL, MongoDB, Docker, Linux, Git
- **Concepts:** REST APIs, JWT Auth, OOP, System Design, Agile/Scrum, DSA

WORK EXPERIENCE

Data Analyst Intern · Genpact – New York, USA *Jun 2025 – Aug 2025*
- Designed and deployed an AI-powered Market Scope Analysis platform using **Agentic AI**, RAG, and LLM-based orchestration; built autonomous agents for web scraping, document retrieval, and vector search, cutting manual research time by ~60%.
- Architected a paired-agent system in CrewAI with retrieval and analysis agents per research area running in parallel, with iterative prompt refinement to **eliminate hallucination**, enforce source-grounded outputs with inline citations, and standardize formatting for consistent report generation.
- Built an AWS Glue ETL pipeline with **dimensional modeling** (star schema) to transform raw data into fact/dimension tables, enabling automated daily analysis of records across multiple business domains.

Summer Intern · EY – Chennai, India *Mar 2024 – Apr 2024*
- Designed an SAP-ERP business blueprint for a simulated retail business across 6 core modules (procurement, planning, manufacturing, sales & distribution); approved by manager for demonstrating end-to-end process understanding.
- Built a multi-module ERP demo application in HTML, CSS, JavaScript, and Node.js, translating the blueprint into functional end-to-end workflows.

Technical Operations Intern · Unfold Consulting – Bangalore, India *Aug 2023 – Nov 2023*
- Executed 250+ UAT test cases across dev and production environments, enabling early go-live on select features.
- Identified a critical defect in the production environment, along with 10+ other database and data-upload issues, preventing potential post-launch failures; retested fixes alongside developers to validate resolution before deployment.

PROJECTS

NutriPlan – Full-Stack AI Meal Planning App Link to code
- Architected a full-stack app using FastAPI, Next.js/TypeScript, and PostgreSQL (SQLAlchemy ORM), with JWT auth via httpOnly cookies across 25+ REST endpoints for auth, preferences, meal generation, pantry, and chat.
- Built a multi-agent system (LangChain + Gemini) separating LLM recipe generation from deterministic validation, including calorie targets (Mifflin-St Jeor), allergy/restriction checks, and a retry loop that reprompts on validation failure.
- Built a conversational agent for intent classification, ambiguity resolution, and direct meal-plan edits, including nutrition recalculation and automatic grocery-list sync.

Document Q&A – Full-Stack RAG Application Link to code Link to Demo
- Architected a containerized full-stack AI application using LangChain agents, ChromaDB, FastAPI, and Next.js/TypeScript, deployed on Railway and Vercel with automated GitHub-triggered redeployments.
- Built FastAPI StreamingResponse endpoints with SSE and a React/TypeScript chat UI featuring token-by-token streaming, PDF upload with live progress tracking, per-document conversation history, and protected routes with token-based session guards.

IoT – Azure Cloud Digital Twin Link to Project 🏆 *Best Innovation, Engenius 2024*
- Built a real-time cattle health monitoring system streaming IoT sensor data through Azure Digital Twins, Azure Functions, and Azure Data Factory, with 3 RESTful API endpoints and an interactive 3D digital twin delivered to a React Native app for veterinarians.

Natural Language Processing – IEEE Published *ieeexplore.ieee.org/document/10544307*
- Fine-tuned Bio+Clinical BERT across three clinical NLP tasks; integrated LIME/SHAP explainability to expose hidden model failures masked by high aggregate accuracy; published in IEEE.